Homework — 2.2.2 Computational Methods — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] — Any two (1 each): can be **decomposed** into sub-problems; has clear **inputs, processes and outputs**; is repetitive / rule-based (algorithmic); allows useful **abstractions**; data can be **structured/modelled**; has a clear, **finite** sequence of steps.

2. [2] — 1 mark: **decomposition** breaks a problem into smaller, manageable sub-problems that may be **different** sub-tasks. 1 mark: **divide and conquer** repeatedly splits a problem into **smaller instances of the *same* problem**, solves the parts, then **combines** the results (e.g. binary search, merge sort). *(Key contrast: different sub-tasks vs same-type smaller instances combined.)*

3. - (a) **[1]** — **Visualisation**. - (b) **[1]** — Any one: patterns/trends/correlations/outliers are easier to spot than in raw figures; aids faster decision-making; communicates findings more clearly.

4. - (a) **[1]** — A **heuristic** is a rule of thumb that gives a **good-enough (not guaranteed optimal)** answer quickly. - (b) **[2]** — 1 mark: an exact method must check an enormous number of route combinations — the work grows factorially, so it is **intractable** / too slow for many locations. 1 mark: a heuristic finds a **good-enough** route **quickly**, accepting that it may **not be optimal**, which is acceptable in practice.

5. [2] — 1 mark: pipelining splits a process into **stages** so that different stages can work on **different items at the same time**, increasing throughput. 1 mark: valid example, e.g. CPU **fetch–decode–execute** (while one instruction is decoded, the next is fetched); or an assembly line.

6. [3] — 1 mark: **representational abstraction** removes/hides detail not relevant to the problem, keeping only what is needed. 1 mark: **abstraction by generalisation** groups things by **shared characteristics** so they can be treated the same. 1 mark: a valid example of **each** — e.g. the London Tube map (keeps connections, drops true distances) and classifying shapes as "polygon" / car types as "vehicle".

Part B (6 marks)

7. [1] — A **function returns a single value** (usable in an expression); a procedure returns no value.

8. [2] — 1 mark: **stack overflow**. 1 mark: with no reachable base case the subroutine keeps calling itself, each call **pushing a new stack frame**, until memory is exhausted.

9. [3] — 1 mark: **thinking ahead** = identifying in advance the inputs, outputs and reusable components a solution will need / anticipating needs (e.g. preconditions, caching) before or while designing it. 1 mark per distinct example, max 2 — e.g. determining what data a form must capture before building it; **caching** likely-needed web resources; pre-fetching in a CPU pipeline; identifying preconditions for a subroutine.

Total: 14 + 6 = 20